

Adaptive AI-Driven Intrusion Detection & Protection System using Federated Learning (A-IDAPS-FL)

Phase 2 Project Report

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Abstract

Phase 1 of A-IDAPS-FL delivered a centralized hybrid intrusion detection system (Autoencoder + XGBoost) trained on the DDoS day of CIC-IDS2017 and served through a real-time Streamlit dashboard. Phase 2 turns the system into a privacy-preserving federated one. Four simulated network nodes, each seeing a different mix of attacks, train the models locally with the Flower framework and exchange only model parameters over TLS; a central server aggregates them (FedAvg for the Autoencoder, tree bagging for XGBoost), calibrates the decision thresholds on its own labelled hold-out and publishes a global model. The training data was widened to all eight CIC-IDS2017 capture days (2.83 million flows, seven attack families). On a 283,074-flow report split the federated hybrid model reaches 99.23% accuracy, 99.95% recall and 0.95% false-positive rate, against 99.55% / 99.68% / 0.49% for a centralized model that needed 2.26 million raw flow records uploaded to the server, and 88.92% / 51.75% / 1.96% for the Phase 1 model. Phase 2 also adds autonomous prevention (host-firewall blocking with audit and undo), per-prediction explainability (TreeSHAP and a LIME-style local surrogate), persistent event logging, a differential-privacy option with a formal (ϵ, δ) analysis, an LSTM temporal-model experiment, CSE-CIC-IDS2018 schema support and a regression test suite. During Phase 2 several defects of the Phase 1 deployment were found and corrected; they are documented in Chapter 3 because they change how the Phase 1 results must be read.

Chapter 1 — Introduction

1.1 Motivation

Signature-based intrusion detection cannot see new attacks, machine-learning detection needs large labelled traffic sets, and collecting that traffic centrally is itself a privacy and compliance problem. Federated learning (McMahan et al., 2017) trains one model across many data owners without moving the data. Phase 2 applies it to the hybrid detector built in Phase 1.

1.2 Problem statement

Build a federated version of the A-IDAPS-FL detector in which (a) raw traffic never leaves a node, (b) the global model detects every attack family present in CIC-IDS2017, (c) transport is encrypted, (d) detection is explainable per flow, (e) confirmed attacks can be blocked automatically, and (f) the privacy, accuracy and communication trade-offs are measured, not asserted.

1.3 Scope of Phase 2 (10-week plan)

Federated learning study and Flower setup; TLS between nodes and server; multi-attack dataset preparation over four non-IID nodes; local Autoencoder and XGBoost training; FedAvg / bagging aggregation; differential privacy; convergence and communication-overhead measurement; centralized vs federated comparison; dashboard update; final testing and documentation.

Chapter 2 — Literature Survey (additions to Phase 1)

- **Federated averaging.** McMahan et al. (2017) average client weight updates weighted by local sample count. It applies directly to the Autoencoder.
- **Federated gradient-boosted trees.** Trees have no weights to average. Flower's `FedXgbBagging` strategy (Flower Labs, 2023) lets every client grow trees on top of the shared booster and appends them; we use it for XGBoost.
- **Differential privacy in FL.** McMahan et al. (2018) clip client updates and add Gaussian noise at the server (client-level DP); the privacy budget is accounted with Rényi DP (Mironov, 2017). DP-FedAvg is designed for hundreds to thousands of clients.
- **Explainability.** SHAP (Lundberg & Lee, 2017) via TreeSHAP for the XGBoost component, and LIME (Ribeiro et al., 2016) as a model-agnostic local surrogate of the whole hybrid rule.
- **Dataset.** CIC-IDS2017 (Sharafaldin et al., 2018); CSE-CIC-IDS2018 uses the same CICFlowMeter features under abbreviated names.

Chapter 3 — Audit of the Phase 1 Deployment

Before federating the Phase 1 models we replayed the CIC-IDS2017 data through the deployed code. The findings below were fixed in Phase 2 and must be kept in mind when citing Phase 1 numbers.

Finding	Effect	Fix
<code>threshold.json</code> held 0.5 (the notebook's ensemble cutoff) instead of the Autoencoder reconstruction threshold 1.149e-4	Deployed predictor had 0% attack recall on its own training day	Threshold file now carries the calibrated reconstruction threshold and the classifier cutoff
Live extractor used seconds and Ethernet frame lengths; the dataset uses microseconds and payload bytes; backward window never filled	Live traffic never resembled training data, which is why Phase 1 needed an XGBoost "gate" that made live detection impossible	Extractor rewritten to CICFlowMeter conventions; gate removed; unit tests
Model trained on the DDoS day only	On multi-attack data: 5% recall on port scans, 0.1% on brute force	Retrained on all eight days

Finding	Effect	Fix
Dashboard latency was <code>random.uniform(6, 20)</code> , confusion matrix guessed from scores, accuracy hard-coded, SHAP table static	Displayed metrics were not measurements	Measured latency, ground-truth-only confusion matrix, hold-out accuracy, per-prediction SHAP
The 99% "hybrid" accuracy in the Phase 1 notebook came almost entirely from XGBoost (mean DL contribution 0.018)	The Autoencoder was not the primary decision maker as documented	Decision rule redesigned (Chapter 5)

Chapter 4 — Dataset and Federated Partitioning

4.1 Data

All eight `MachineLearningCVE` files of CIC-IDS2017: 2,830,743 flows after cleaning (no rows lost). Labels are mapped to families: BENIGN 2,273,097; DoS 252,661 (Hulk, GoldenEye, slowloris, Slowhttptest); PortScan 158,930; DDoS 128,027; BruteForce 13,835 (FTP-Patator, SSH-Patator); WebAttack 2,180 (brute force, XSS, SQL injection); Bot 1,966; Other 47 (Infiltration, Heartbleed). The 30 features selected in Phase 1 are kept so results remain comparable. One `MinMaxScaler` is fitted on all rows.

4.2 Hold-out and its two halves

A stratified 20% (566,149 flows) is held by the server and split deterministically: even rows form the **calibration** half (thresholds), odd rows the **report** half (every number in this report). Nothing tuned on one half is measured on it.

4.3 Non-IID nodes

Each node models an organisation that mostly sees one kind of attack plus a random share of benign traffic.

Node	Attack families	Rows	Attacks
0	DDoS, DoS	668,245	304,550
1	PortScan, Bot	492,412	128,717
2	BruteForce, WebAttack	376,508	12,812
3	almost only benign (Other)	727,429	38

Chapter 5 — Hybrid Decision Rule

```
dl_score = clip( mse / (2 * 10 * ae_threshold), 0, 1 )      0.5 at 10× the benign 90th
percentile
ml_score = piecewise-linear rescaling of P(attack) so that ml_score = 0.5 at the
calibrated cutoff
```

```
final_score = max(dl_score, ml_score)
0.5
```

```
ATTACK if final_score >
```

A flow is an attack when **either** the supervised model recognises a known signature **or** the reconstruction error is far above anything benign traffic produces. Candidate rules were compared on the DDoS and PortScan days: the Phase 1 additive rule gave 89.7% accuracy at 18.8% FPR; averaging gave 97.6%; the max rule with a 10× anomaly multiplier gave 99.8% at 0.5% FPR. The multiplier and the classifier cutoff are the only tuned scalars and both are chosen on the calibration half.

Chapter 6 — Federated Architecture

6.1 Components

Component	Implementation
Framework	Flower 1.36, gRPC deployment mode (<code>start_server</code> / <code>start_client</code>), one process per node
Transport security	TLS: self-signed root CA + server certificate (<code>federated/gen_certs.py</code>); clients verify the CA
Autoencoder aggregation	FedAvg weighted by local benign rows; 15 rounds, 1 local epoch, batch 1024
XGBoost aggregation	Bagging: each node adds 3 parallel trees per round on top of the global booster, the server appends them; 40 rounds → 480 trees; eta 0.1, depth 8, per-node <code>scale_pos_weight</code>
Server-side evaluation	Every round on the report half; per-round JSON logs
Calibration	<code>federated/calibrate.py</code> : AE threshold = benign 90th percentile; XGBoost cutoff = best F1 under 1% FPR, both on the calibration half
Differential privacy	Flower <code>DifferentialPrivacyServerSideFixedClipping</code> (client-level, fixed clipping + Gaussian noise), optional
Export	Global model, scaler, thresholds and per-family flow prototypes installed into <code>hybrid_model/</code> ; Phase 1 artifacts backed up

6.2 Why calibration is required under non-IID bagging

Nodes 2 and 3 hold almost no attacks, so the trees they add every round push probabilities down. The global booster ranks flows almost perfectly (AUC 0.9998 from round 1) but at a fixed 0.5 cutoff its recall is only 27% after 40 rounds. Calibrating the cutoff on the server's labelled calibration split (chosen value 0.164) restores 99.95% recall at 0.25% FPR. This is a property of federated bagging on heterogeneous data, and the calibration step is part of the design.

6.3 Data flow

```
node k: local flows → scale → train AE epoch / grow XGB trees → parameters or new trees
—TLS→ server
```

server: aggregate → evaluate on report half → log round → next round
final : calibrate on calib half → save global model → export to dashboard

Chapter 7 — Results

7.1 Detection performance (report split: 283,074 flows, 55,741 attacks)

Model	Detector	Accuracy	Precision	Recall	F1	FPR
Phase 1 (DDoS day)	Autoencoder	85.60%	81.45%	34.77%	48.73%	1.94%
Phase 1 (DDoS day)	XGBoost	84.87%	99.67%	23.24%	37.69%	0.02%
Phase 1 (DDoS day)	Hybrid	88.92%	86.62%	51.75%	64.79%	1.96%
Phase 2 centralized	Autoencoder	86.07%	94.76%	30.95%	46.66%	0.42%
Phase 2 centralized	XGBoost	99.88%	99.72%	99.68%	99.70%	0.07%
Phase 2 centralized	Hybrid	99.55%	98.04%	99.68%	98.86%	0.49%
Phase 2 federated	Autoencoder	85.33%	91.00%	28.28%	43.15%	0.69%
Phase 2 federated	XGBoost	99.77%	98.92%	99.94%	99.43%	0.27%
Phase 2 federated	Hybrid	99.23%	96.27%	99.95%	98.07%	0.95%

Recall per attack family, hybrid rule:

Family	Phase 1	Centralized	Federated
DDoS	99.99%	99.97%	99.97%
DoS	60.49%	99.93%	99.92%
PortScan	5.27%	100%	100%
BruteForce	0.14%	99.93%	100%
Bot	5.94%	66.83%	97.52%
WebAttack	3.72%	60.00%	99.53%

The federated model loses 0.3 points of accuracy to the centralized one (a higher false-positive rate from the autoencoder component) but detects the two rare families better because per-node `scale_pos_weight` and the calibrated cutoff give minority attacks more weight than one global weight does.

7.2 Convergence

- Autoencoder: benign reconstruction error falls from $2.4e-1$ to $2.8e-5$ over 15 rounds; recall reaches 99% of its final value at round 4.
- XGBoost: AUC 0.9998 at round 1 and flat thereafter; logloss keeps falling to 0.171 at round 40.

7.3 Communication overhead

Quantity	Value
Centralized: raw rows that must be uploaded	2,264,594 flows = 274 MB
Federated Autoencoder	117 KB per node per round; 933 KB per round; 14.0 MB for 15 rounds
Federated XGBoost	256.6 MB for 40 rounds (the growing 3.1 MB global booster is redistributed to 4 nodes every round)
Federated total	270.6 MB, i.e. $0.99 \times$ the raw upload, but consisting only of parameters and trees

The XGBoost traffic is dominated by redistributing the full booster; sending only the trees added since the previous round would cut it to roughly 12 MB. This optimisation is noted as future work.

7.4 Differential privacy

Client-level (ϵ, δ) -DP with $\delta = 1e-5$, Rényi accounting of the Gaussian mechanism, 4 clients participating every round (noise std per weight = $z \cdot C / 4$):

Setting	ϵ	Benign MSE	AE recall	Usable
no DP	∞	2.8e-5	0.282	reference
$z = 0.3$, $C = 1.0$, 15 rounds	145	1.5e-1	0	no
$z = 0.05$, $C = 20$, 15 rounds	3,372	1.9e-1	0	no
$z = 0.01$, $C = 1.0$, 15 rounds	76,901	9.6e-4	0.005	no
$z = 0.01$, $C = 1.0$, 40 rounds	203,151	3.3e-5	0.300	accuracy yes, privacy no

Conclusion. With four clients the noise required for a meaningful ϵ (below 10) is larger than the model weights themselves, and the noise level at which the Autoencoder still trains gives an ϵ that offers no privacy. DP-FedAvg needs the noise to average over hundreds of clients. In this deployment privacy therefore rests on the federated architecture itself (raw traffic never leaves a node) and on TLS, not on differential privacy. The pipeline supports DP end to end so the experiment can be repeated with more simulated nodes.

7.5 Temporal model (LSTM) experiment

Windows of 10 consecutive flows (chronological order within each capture day), chronological 70/30 split per day, window labelled ATTACK if any flow is an attack; LSTM(64) $\rightarrow$ Dense(32) $\rightarrow$ sigmoid, 4 epochs, class-weighted.

Detector (window level)	Accuracy	Precision	Recall	F1	FPR
LSTM over 10-flow windows	88.91%	78.96%	62.69%	69.89%	4.31%
Federated hybrid, any flow in window	95.93%	83.92%	99.16%	90.90%	4.91%

The LSTM matches the hybrid on DDoS (99.4%) and PortScan (96.9%) but fails on DoS (0.3%), BruteForce (1.4%) and Bot (23.9%): under the chronological split those families appear as different sub-attacks in the training and test portions of a day (for example FTP-Patator in the morning, SSH-Patator in the afternoon), so the sequence

model is asked to generalise to variants it never saw, which the flow-level model handles through its shared feature representation. A sequence model is therefore not adopted for Phase 2 production; the experiment is kept as a documented negative result with a reproducible script.

7.6 Live sniffer

The corrected extractor passes six unit tests on synthetic packets (units, direction, payload lengths, UDP window convention, bounded cache). Single-flow inference latency measured in the dashboard is 30–40 ms cold and about 2 ms warm. Capture on a real interface requires Npcap and Administrator rights and was not part of the automated tests.

Chapter 8 — Protection: Autonomous Prevention

`core/prevention.py` implements the "Protection" half of the project name.

- **Policy.** A source is blocked only when its flows scored ≥ 0.75 at least 3 times within 60 s. Loopback, link-local, multicast, private (LAN) addresses, the host's own addresses, the simulator and an allowlist are never blocked.
 - **Backends.** Windows `netsh advfirewall` rule per IP; Linux `iptables` DROP rule; **dry-run** (decide and log only) is the default and is what the dashboard toggle enables first.
 - **Audit and undo.** Every skip, block and unblock is appended to `logs/prevention.jsonl` and to the SQLite event store; blocked IPs persist across restarts and can be unblocked from the dashboard.
 - **Tests.** The policy is unit-tested in dry-run mode (thresholds, hit counting, exclusions, undo).
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Chapter 9 — Explainability and Persistence

- **TreeSHAP** (exact, via XGBoost `pred_contribs`) shows which features pushed the classifier toward ATTACK for the specific flow.
 - **Autoencoder error share** shows which features the anomaly detector could not reconstruct.
 - **LIME-style local surrogate** perturbs the flow 500 times, weights samples by proximity and fits a ridge regression to the *final hybrid score*, giving a model-agnostic view of the whole decision, with the local fit R^2 reported.
 - **Event store.** `logs/ids_events.sqlite` keeps every verdict, alert and prevention action, closing the Phase 1 "no persistent logging" limitation.
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Chapter 10 — Dashboard (Phase 2)

Measured latency; ground-truth-only confusion matrix (simulated flows carry their true label, live flows are counted separately); hold-out accuracy from `model_info.json`; attack simulator seeded from the median real flow of each family with optional knob overrides; per-flow XAI panel with LIME toggle; autonomous-prevention panel (enable, apply-to-firewall, blocked list with unblock); federated-training panel (node partitions, per-round

recall/FPR curves, traffic, TLS/DP flags, DP and LSTM tables, centralized-vs-federated table). Verified with Streamlit's `AppTest` harness end to end.

Chapter 11 — Testing

Suite	Tests	Covers
<code>tests/test_features.py</code>	6	extractor units, direction, payload lengths, UDP, cache bound
<code>tests/test_predictor_regression.py</code>	5	pure decision rule, threshold-file sanity, replay of 20,000 real flows (recall ≥ 0.90 , FPR ≤ 0.05 , per-family floors), single vs batch API
<code>tests/test_phase2_modules.py</code>	5	prevention policy (dry-run), DP accounting monotonicity, event store, LIME shape

All 16 tests pass against the exported federated model. The replay regression is the guard that would have caught the Phase 1 threshold defect.

Chapter 12 — Conclusion and Future Work

Phase 2 delivers a federated, TLS-secured, explainable and self-protecting IDS whose global model detects all CIC-IDS2017 attack families at 99.23% accuracy and 99.95% recall without any raw flow leaving a node, at a communication cost comparable to a one-time raw upload. Two negative results are reported honestly: differential privacy is not meaningful with four clients, and a 10-flow LSTM does not beat the flow-level hybrid under a strict chronological split.

Future work: (1) send only per-round tree deltas to cut XGBoost traffic by $\sim 20\times$; (2) repeat the DP study with 50–200 simulated clients; (3) evaluate on CSE-CIC-IDS2018 with the schema mapping already in `prepare_data.py`; (4) federate the LSTM and test on longer, source-grouped sequences; (5) client authentication keys in addition to server TLS; (6) live validation against controlled attack traffic in an isolated lab network.

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